

CLIMATE CHANGE TREND ANALYSIS AND FORECASTING

Haindavi Paladugu,
B.Tech in Mathematics and Computing (2024-2028)
Central University of Karnataka

Mentor Name: Sauparna Sarkar

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1. ABSTRACT

This project focuses on historical analysis and future forecasting of global greenhouse gas emissions. The study uses the Our World in Data (OWID) CO₂ dataset to analyze trends in carbon dioxide and other greenhouse gas emissions across different countries and time periods. Exploratory data analysis is performed to identify emission patterns, major contributors, and long-term climate trends. Various machine learning and statistical forecasting models, including Linear Regression, Random Forest, and Exponential Smoothing (ETS), are implemented to predict future emission levels. The forecasting results provide insights into future emission trajectories and uncertainty trends. An interactive Streamlit dashboard is developed to visualize historical patterns, country-wise comparisons, and forecasting outcomes. The project demonstrates the application of data analytics, machine learning, and time-series forecasting techniques for understanding global climate change trends.

2. INTRODUCTION

Climate change is one of the major global challenges, primarily due to the increasing concentration of greenhouse gases (GHGs) in the atmosphere due to human activities such as industrialization, energy production, etc. Analyzing greenhouse gas emissions is essential for understanding climate trends, identifying major emission contributors, and developing strategies for sustainable environmental. This project focuses on historical analysis and future forecasting of global greenhouse gas emissions.

The project utilizes the Our World in Data (OWID) CO₂ dataset, which provides historical records of carbon dioxide emissions and other greenhouse gas indicators across countries and years. Exploratory Data Analysis (EDA) techniques are applied to understand emission patterns, compare country-wise contributions, and identify long-term trends.

The project involves data preprocessing, feature engineering, visualization, implementation of predictive models, and performance evaluation. Machine learning models such as Linear Regression and Random Forest, along with statistical forecasting methods like Exponential Smoothing (ETS), are used to predict future emission levels. Model evaluation is performed using metrics such as Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). An interactive Streamlit dashboard is developed to provide a visualization platform for exploring historical trends, country comparisons, and forecast results.

The primary purpose of this project is to apply data-driven techniques to analyze greenhouse gas emission patterns and provide insights into future emission behavior.

During the initial two weeks of the internship, training was provided on the following topics:

- Introduction to Data Science and Machine Learning
- Python Programming
- Data Analysis using NumPy and Pandas
- Exploratory Data Analysis (EDA)

- Data Visualization
- Machine Learning Fundamentals
- Time-Series Analysis and Forecasting

3. PROJECT OBJECTIVE

- To analyze historical greenhouse gas emission trends using the Our World in Data (OWID) CO₂ dataset.
- To identify emission patterns and compare greenhouse gas emissions across different countries over time.
- To develop forecasting models using Linear Regression, Random Forest, and Exponential Smoothing (ETS) to predict future CO₂ emissions.
- To evaluate and compare the performance of different forecasting models using MAE and RMSE.
- To develop an interactive Streamlit dashboard for visualizing historical trends, country-wise analysis, and future emission forecasts.

4. METHODOLOGY

The methodology adopted for the project is described as follows:

DATA COLLECTION:

The dataset used for this project was obtained from the **Our World in Data (OWID) CO₂ Dataset**, which contains historical greenhouse gas emission data for countries across the world. The dataset includes variables such as CO₂ emissions, CO₂ emissions per capita, methane emissions, nitrous oxide emissions, total greenhouse gas emissions, GDP, population, and other climate-related indicators. Only records from 1990 onwards were retained because data coverage is more complete and consistent from this period, and 1990 serves as a common baseline year in international climate policy.

DATA CLEANING & PRE-PROCESSING:

Data cleaning was done before analysis. Duplicate records and unnecessary columns were removed, and missing values were handled appropriately. Aggregate entries such as World, Asia, and Europe were removed to avoid double-counting and ensure that the analysis was performed only on sovereign countries. Data from **1990 onwards (years $\geq$ 1990)** was retained. Variables such as country, year, CO₂ emissions, CO₂ per capita, methane, nitrous oxide, and total greenhouse gas emissions were retained. Additional features were created to improve forecasting performance, including lag features (previous years' emissions), five-year rolling average, year-over-year percentage change, and greenhouse gas intensity.

EXPLORATORY DATA ANALYSIS (EDA):

Exploratory Data Analysis was performed to understand the characteristics of the dataset and identify trends. Historical global CO₂ emission trends were analyzed, and comparisons were made between major emitting countries. Country-wise emission patterns, greenhouse gas distributions, and long-term trends were visualized. Line charts, bar charts, stacked area charts, and scatter plots were created.

Three different forecasting models were implemented and compared:

- Linear Regression
- Random Forest Regression
- Exponential Smoothing (ETS)

Linear Regression was used as a simple baseline model, Random Forest Regression was applied to capture complex non-linear relationships in the data, and the ETS model was used for time-series forecasting by considering the trend present in historical emission values.

MODEL TRAINING:

A time-based train-test split was used instead of a random split to prevent future data from being used during model training, thereby avoiding data leakage and ensuring a realistic forecasting evaluation. Historical data from **1990 to 2018** was used for training, while data from **2019 to 2023** was for testing. This approach ensures that future information is not used while training the models. The forecasting performance of each model was evaluated using **the Mean Absolute Error (MAE)** and **the Root Mean Square Error (RMSE)**. The results of all models were compared to determine their prediction accuracy.

After model evaluation, future CO₂ emissions were forecast for the period **2024 to 2043**. Forecasts were generated for selected countries using the developed models. For the ETS model, confidence intervals were also calculated to represent the uncertainty associated with long-term predictions. These forecasts help illustrate possible future emission trends based on historical data.

An interactive web dashboard was developed using **Streamlit** to present the project results in a user-friendly manner. The dashboard includes historical emission trends, country-wise profiles, emission statistics, and future forecasts. Interactive visualizations were created using **Plotly Express**, allowing users to explore greenhouse gas emission patterns and compare different countries.

TOOLS AND TECHNOLOGIES USED:

- Python
- Visual Studio Code (VS Code)
- Pandas
- NumPy
- Scikit-learn

- Statsmodels
- Plotly Express
- Streamlit
- GitHub

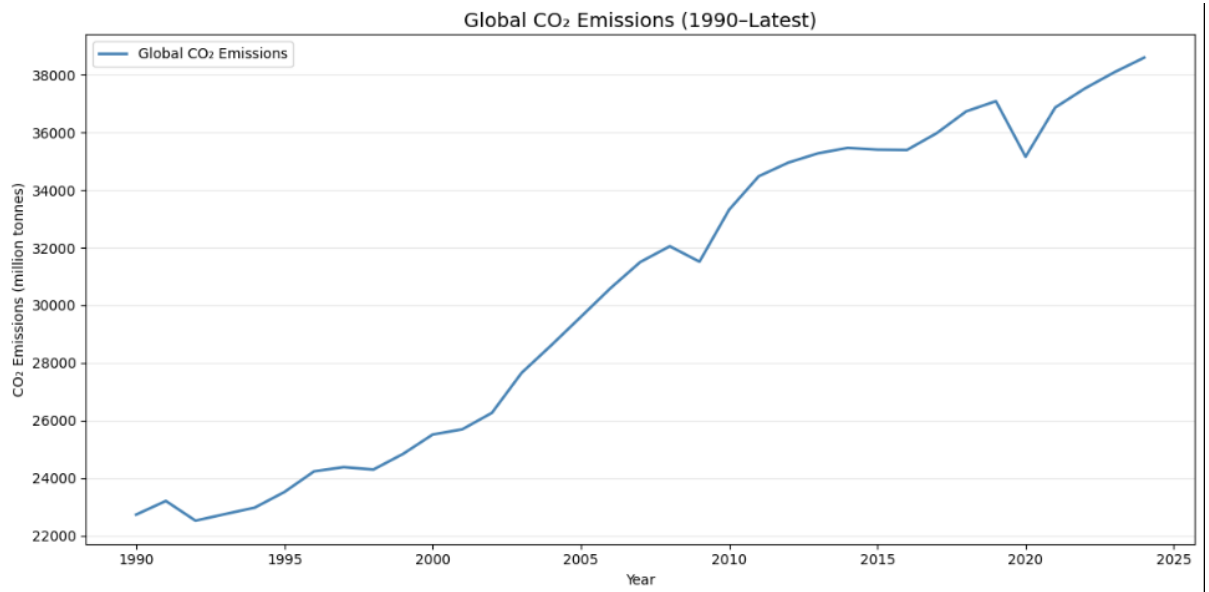
5. DATA ANALYSIS AND RESULTS

The historical greenhouse gas emission dataset was analyzed to understand global emission trends, identify major contributors, and study country-wise emission patterns. Exploratory Data Analysis (EDA) was performed on the processed dataset to extract meaningful insights from historical emission records.

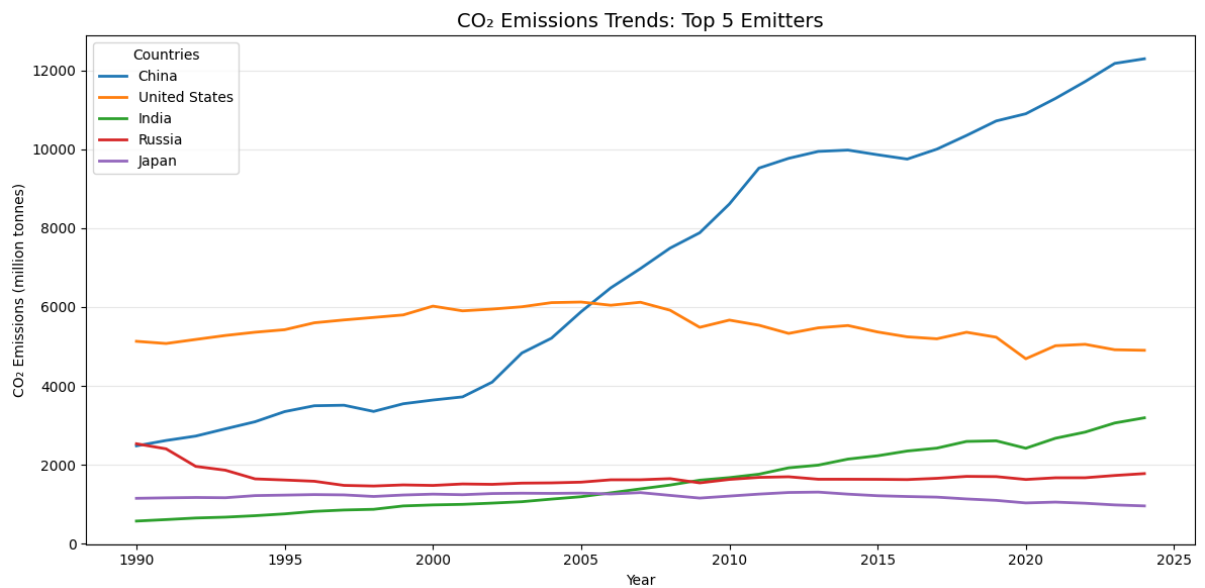
The analysis of global CO₂ emissions from **1990 to 2023** showed an overall increasing trend in emissions, with a temporary decline around 2020 due to reduced industrial and transportation activities during the COVID-19 period. Emissions increased again in the following years. Countries such as China, the United States, and India were identified as major contributors to global CO₂ emissions.

The following visualizations were generated to analyze emission patterns:

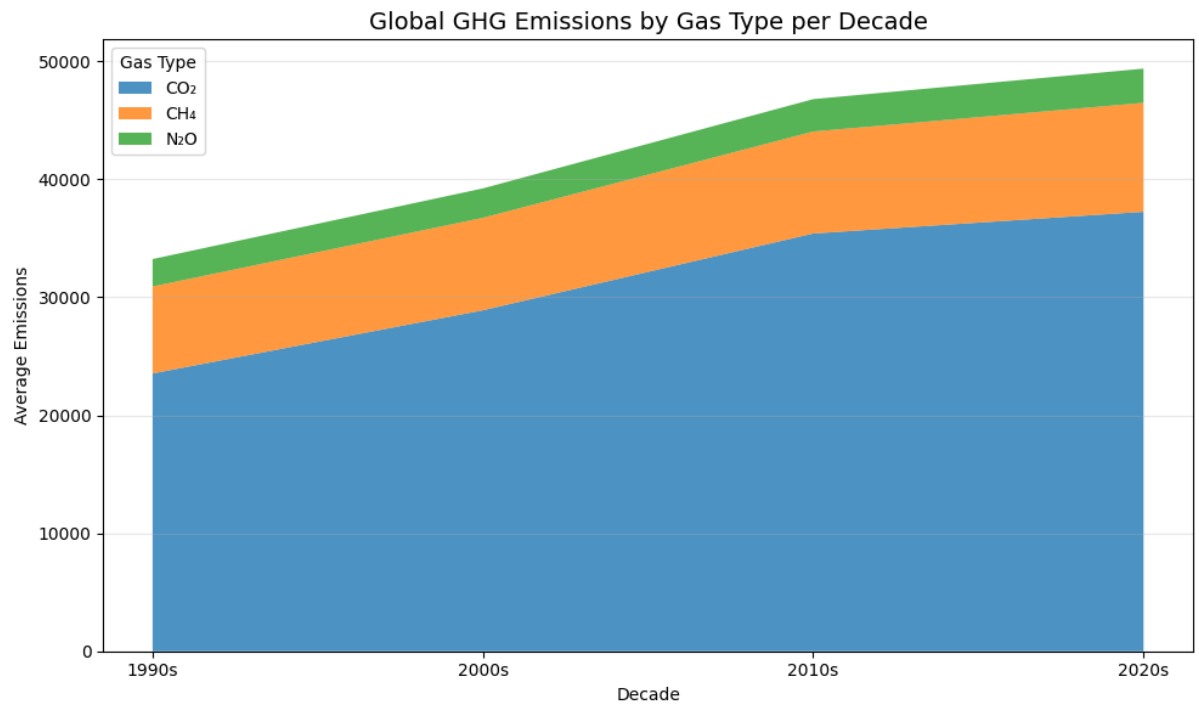
- **Global CO₂ Emissions Trend (1990–2023):** Shows the overall change in global CO₂ emissions over time.



- **Top 5 CO₂ Emitting Countries:** Identifies the countries with the highest contribution to global CO₂ emissions.



- **Greenhouse Gas Composition/Distribution:** Represents the contribution of different greenhouse gases.



Three forecasting models were developed and compared for predicting future CO₂ emission levels:

- Linear Regression
- Random Forest Regression
- Exponential Smoothing (ETS)

The dataset was divided using a time-based train-test split to maintain the chronological order of emission data.

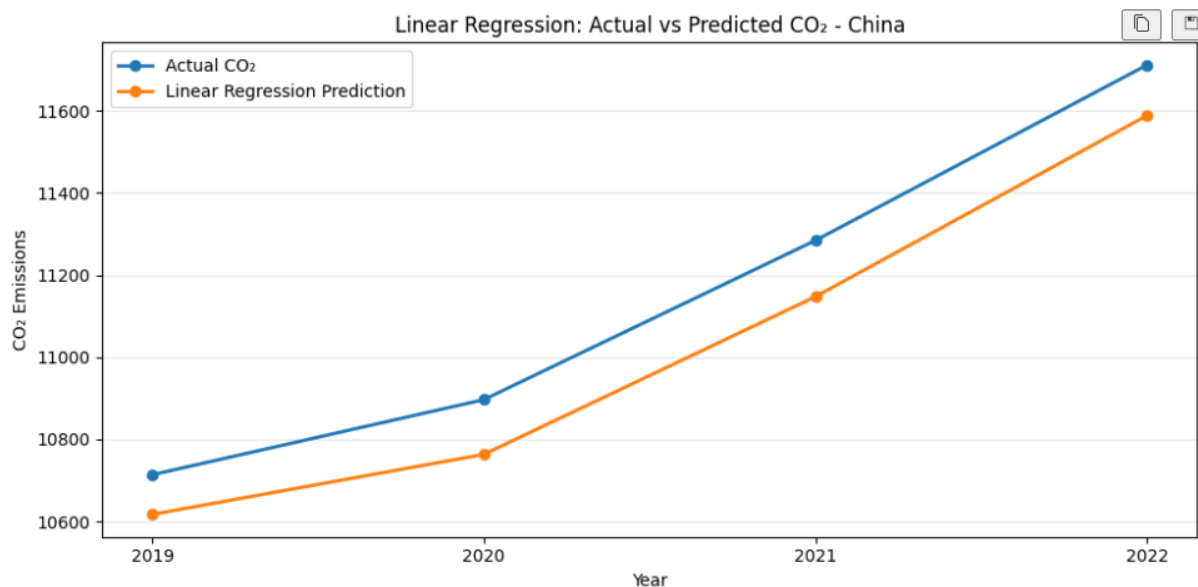
- **Training Data:** 1990–2018

- **Testing Data: 2019–2023**

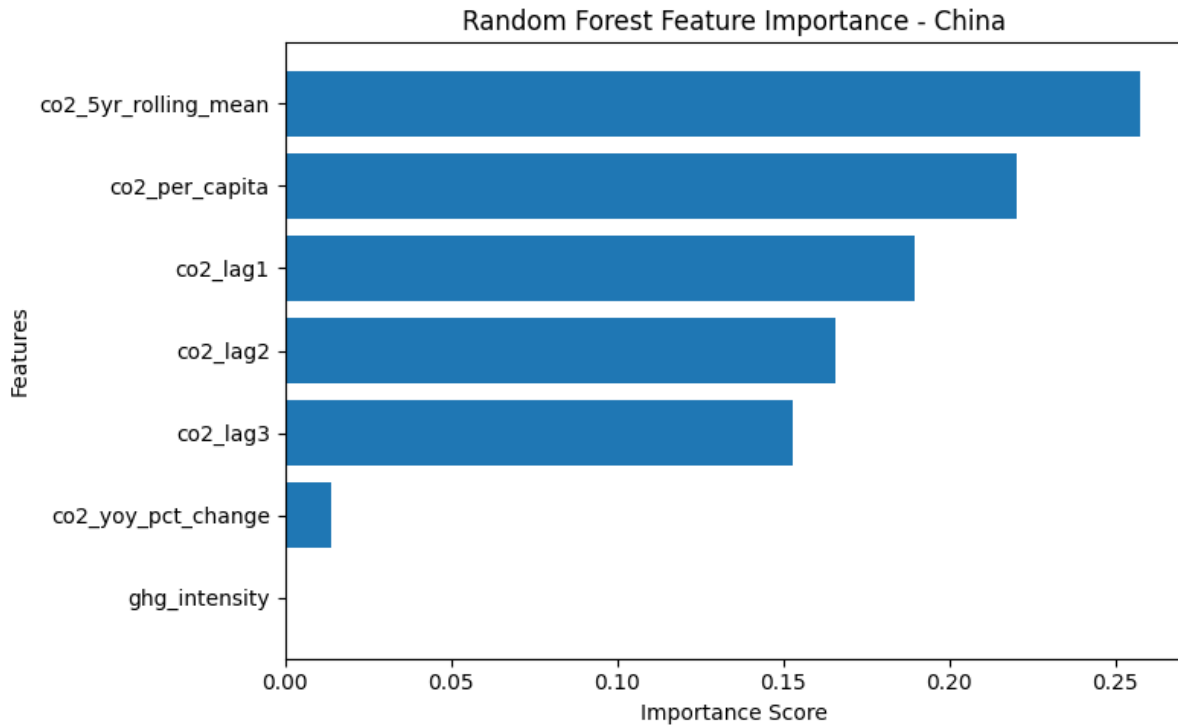
The models were trained using historical emission data and evaluated using prediction accuracy metrics such as **Mean Absolute Error (MAE)** and **Root Mean Square Error (RMSE)**.

	country	LR MAE	LR RMSE	RF MAE	RF RMSE	Best Model
0	China	122.477459	123.491460	964.943285	1037.822098	Linear Regression
1	India	18.558444	21.881422	156.375047	184.781198	Linear Regression
2	Japan	5.161882	5.666255	103.518975	107.144657	Linear Regression
3	United States	33.666872	42.815094	279.267180	341.888192	Linear Regression
4	Russia	3.071546	3.548230	11.808643	13.733550	Linear Regression
5	Germany	7.732684	9.130234	94.110955	96.820607	Linear Regression
6	United Kingdom	6.943559	8.348277	51.632507	56.147720	Linear Regression
7	Brazil	6.385103	6.977136	9.523640	11.830554	Linear Regression
8	Canada	0.423022	0.536373	15.183607	17.397895	Linear Regression
9	Australia	0.494231	0.713844	11.734588	12.999249	Linear Regression

The actual versus predicted emission values were compared to evaluate how effectively each model captured historical trends. The performance comparison helped in selecting the most suitable model for future forecasting.

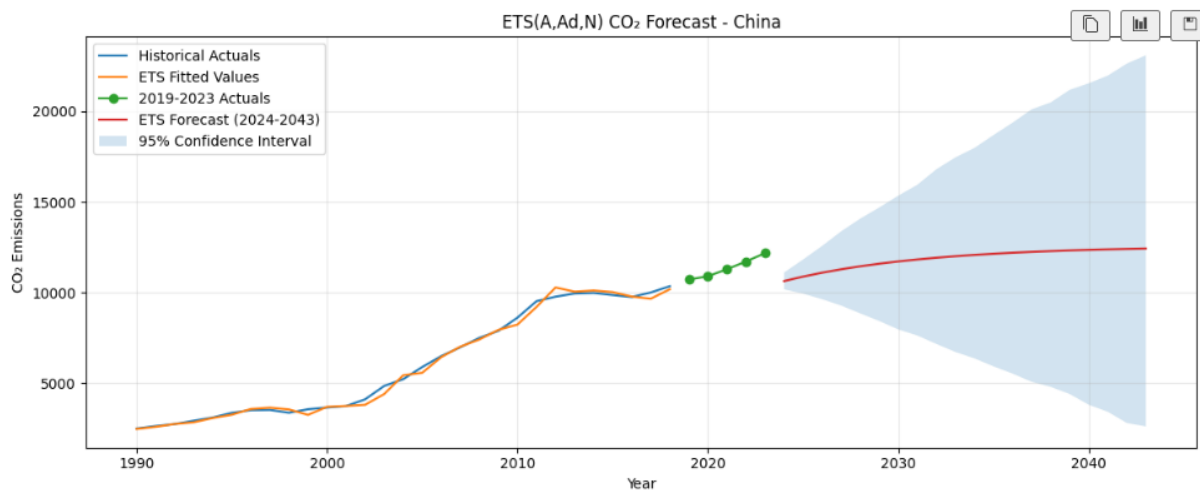


For the Random Forest model, feature importance analysis was performed to understand the contribution of different input variables in predicting CO₂ emissions.



Forecast Results: The selected forecasting models were used to predict future CO₂ emission levels for the period **2024–2043**. The ETS model was applied to capture historical trends and generate future emission forecasts along with confidence intervals.

A forecast plot for **China**, one of the largest global CO₂ emitters, is presented below. The graph shows the historical emission trend, forecasted values, and increasing uncertainty over the forecast horizon. The same forecasting methodology was applied to the other selected countries, and the forecast results were generated accordingly.



The widening confidence interval over time indicates that uncertainty increases for longer-term predictions due to the accumulation of forecasting errors and changes in future emission patterns.

6. CONCLUSION

This project successfully analyzed historical greenhouse gas emission trends and developed forecasting models to predict future CO₂ emission levels using data-driven techniques. The analysis of the OWID CO₂ dataset revealed that global CO₂ emissions have generally increased from 1990 onwards, with major contributions from countries such as China, the United States, and India. The exploratory data analysis helped identify country-wise emission patterns, differences in per capita emissions, and variations in greenhouse gas contributions.

Three forecasting approaches, namely Linear Regression, Random Forest Regression, and Exponential Smoothing (ETS), were implemented and evaluated. The model comparison based on MAE and RMSE showed that **Linear Regression achieved better prediction performance for the selected countries compared to Random Forest Regression**. The ETS model was effective for time-series forecasting and was used to generate future emission predictions with confidence intervals, highlighting the uncertainty associated with long-term forecasts.

The developed Streamlit dashboard provides an interactive platform for exploring historical trends, country-wise comparisons, and future emission forecasts. The project demonstrates how machine learning, statistical forecasting, and data visualization techniques can be applied to understand climate-related data and support environmental analysis.

7. APPENDICES

Appendix A: References

The following resources were referred to during the project:

1. Our World in Data – CO₂ and Greenhouse Gas Emissions Dataset
<https://ourworldindata.org/co2-and-greenhouse-gas-emissions>
2. Kaggle – Introduction to Machine Learning
<https://www.kaggle.com/learn/intro-to-machine-learning>
3. Scikit-learn Documentation – Machine Learning Algorithms and Model Evaluation
<https://scikit-learn.org/>
4. Statsmodels Documentation – Exponential Smoothing Models
<https://www.statsmodels.org/stable/tsa.html>

Appendix B: GitHub Repository

The complete source code developed for the project, including data preprocessing, exploratory data analysis, machine learning models, forecasting implementation, and Streamlit dashboard development, is available at:

GitHub Repository: https://github.com/hain-davi/climate_change_trend_analysis_

Appendix D: Project Documents

The following project-related documents are maintained in the repository:

- Project Report
- Presentation Slides
- Dataset/Processed Dataset
- Source Code Files
- Streamlit Dashboard Application
- Requirements File